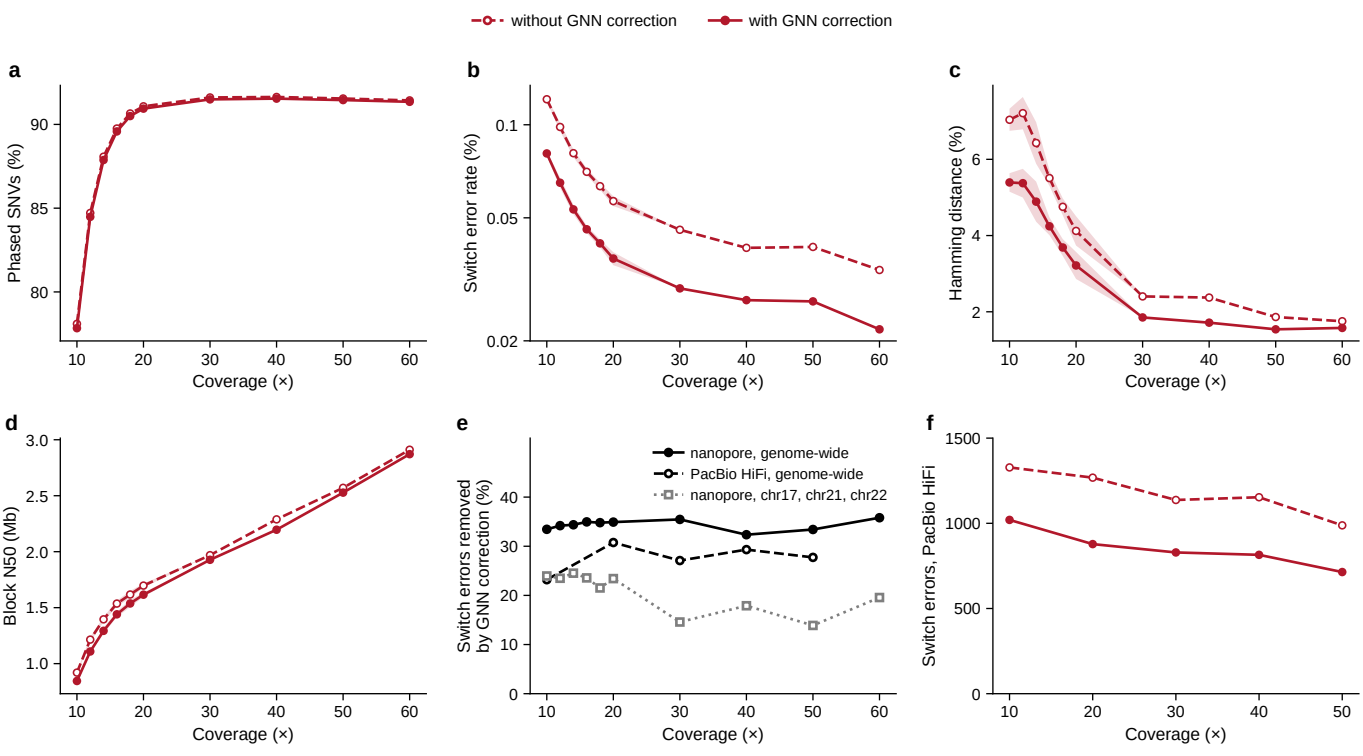
without GNN correction
with GNN correction
a
Phased SNVs (%)
90
85
80
10
20
30
40
50
60
Coverage (×)
b
Switch error rate (%)
0.1
0.05
0.02
10
20
30
40
50
60
Coverage (×)
c
Hamming distance (%)
6
4
2
10
20
30
40
50
60
Coverage (×)
d
Block N50 (Mb)
3.0
2.5
2.0
1.5
1.0
10
20
30
40
50
60
Coverage (×)
e
Switch errors removed by GNN correction (%)
nanopore, genome-wide
PacBio HiFi, genome-wide
nanopore, chr17, chr21, chr22
40
30
20
10
0
10
20
30
40
50
60
Coverage (×)
f
Switch errors, PacBio HiFi
1500
1000
500
0
10
20
30
40
50
Coverage (×)